A constant-source, constant-sink methane trajectory

Methane surge reconstruction

The dashed trajectories in He et al. (2026), Fig. 1, motivate a constant-parameter counterfactual. Holding gross emissions E, soil uptake L (Tg CH_4 yr^{-1}), and chemical loss frequency k (yr^{-1}) fixed gives

$$\frac{dm_c}{dt} = E - L - km_c, \qquad m_c(0) = m_0. \tag{1}$$

For $k > 0$, direct integration gives

$$m_c(t) = m_0 + \left(\frac{E-L}{k} - m_0\right)\left[1 - \exp(-kt)\right]. \tag{2}$$

The limit $k = 0$ is $m_c(t) = m_0 + (E-L)t$. Time is measured in years from the specified initial burden. No feedback multiplier is introduced: holding OH fixed specifies the loss frequency explicitly.

The difference $m_c(t) - m_0$ is a concentration-trajectory attribution. It differs from summing the relaxation term of He et al.'s equation (1), because that term uses the actual m_j, not the counterfactual m_c. Increased emissions also increase subsequent losses, and their effects on burden decay with chemical lifetime. In continuous time, with constant k_0 and imposed forcing $F(t)$, the perturbation is

$$m(t) - m_c(t) = \int_0^t \exp[-k_0(t-s)]F(s)\,ds. \tag{3}$$

We will report the published annual budget decomposition and the constant-parameter trajectory separately. Treating them as numerically identical would introduce an additional approximation.